Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 100 epochs with a batch size of 64. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy of MLP by dataset.

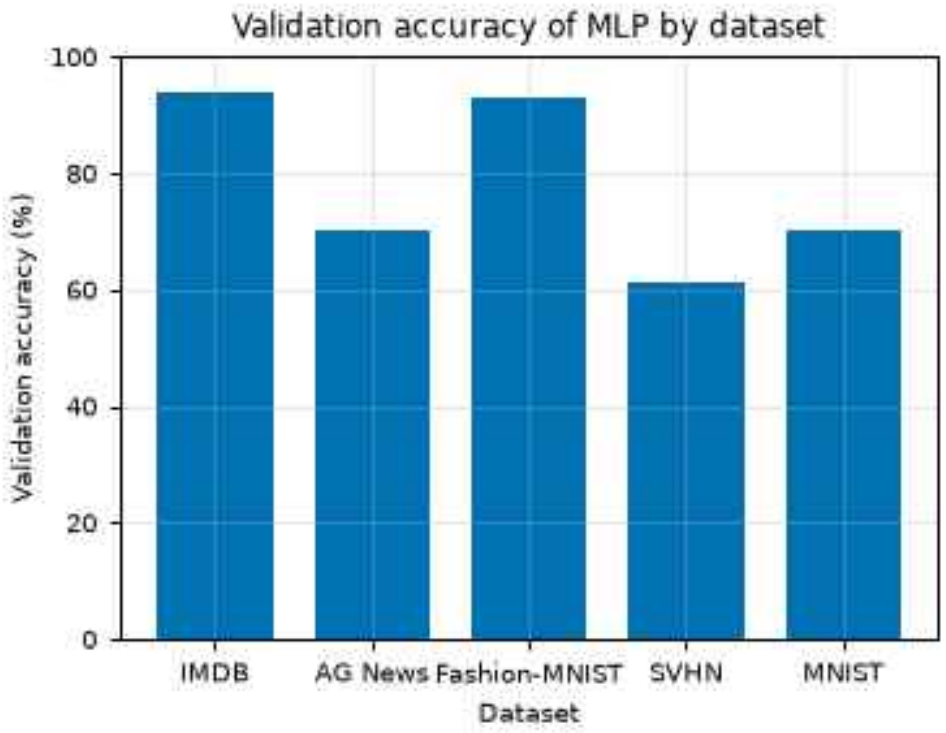


Figure 1: Validation accuracy of MLP by dataset for the models in this experiment.

As shown in Figure 1, MLP posts 72.3% validation accuracy when tested on AG News. MLP posts 70.0% validation accuracy when tested on MNIST.

Figure 2 shows the validation accuracy by dataset.

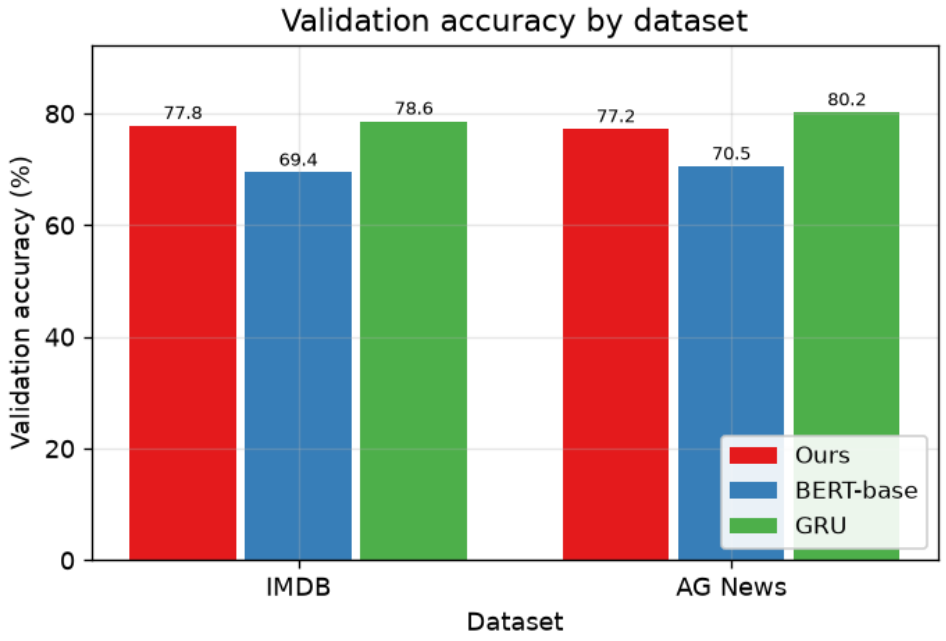


Figure 2: Validation accuracy by dataset for the models in this experiment.

Evaluated on AG News, GRU comes in at 84.8% validation accuracy. Evaluated on AG News, GRU comes in at 71.0% validation accuracy. On IMDB, Ours obtains a validation accuracy of 31.7%.

Figure 3 shows the validation accuracy by dataset.

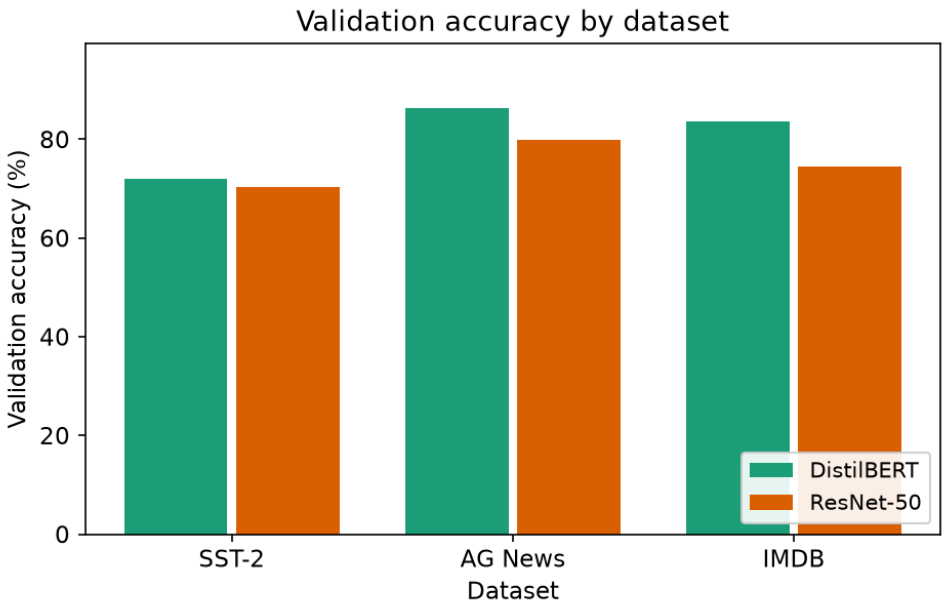


Figure 3: Validation accuracy by dataset for the models in this experiment.

On SST-2, ResNet-50 obtains a validation accuracy of 60.4%. On AG News, DistilBERT is 4.6 points better than ResNet-50.

Figure 4 shows the test accuracy during training.

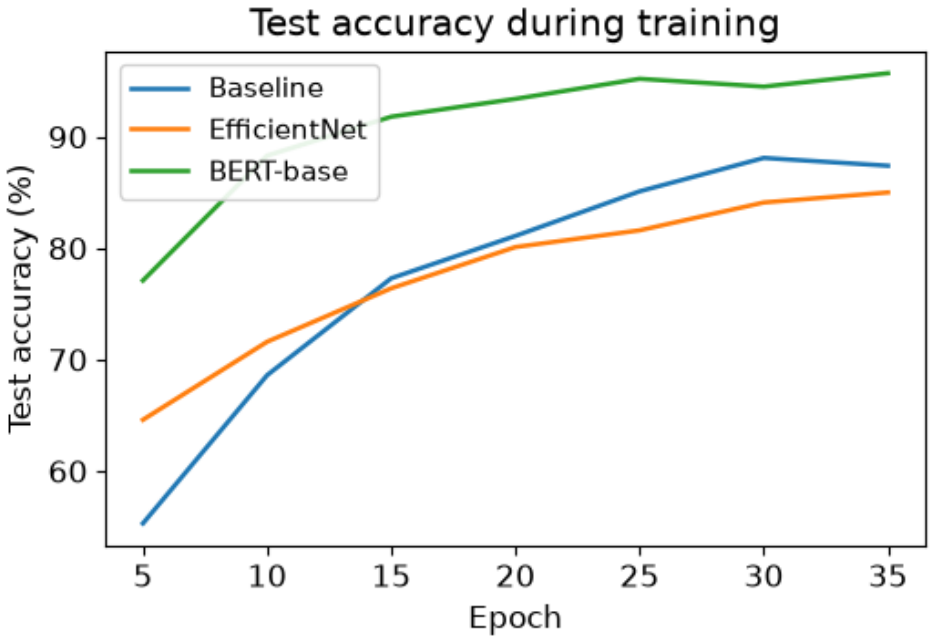


Figure 4: Test accuracy during training for the models in this experiment.

The test accuracy of EfficientNet improves by a relative 11% between epoch 5 and epoch 10. As shown in Figure 4, The test accuracy of BERT-base improves by a relative 14.5% between epoch 5 and epoch 10. Epoch 15 sees Baseline post a test accuracy of 77.3%. By epoch 10, the test accuracy of BERT-base stands at 79.4%.

3 Discussion

On MNIST, Logistic Regression obtains an accuracy of 60.0%. The differences between the models are discussed in the next section.